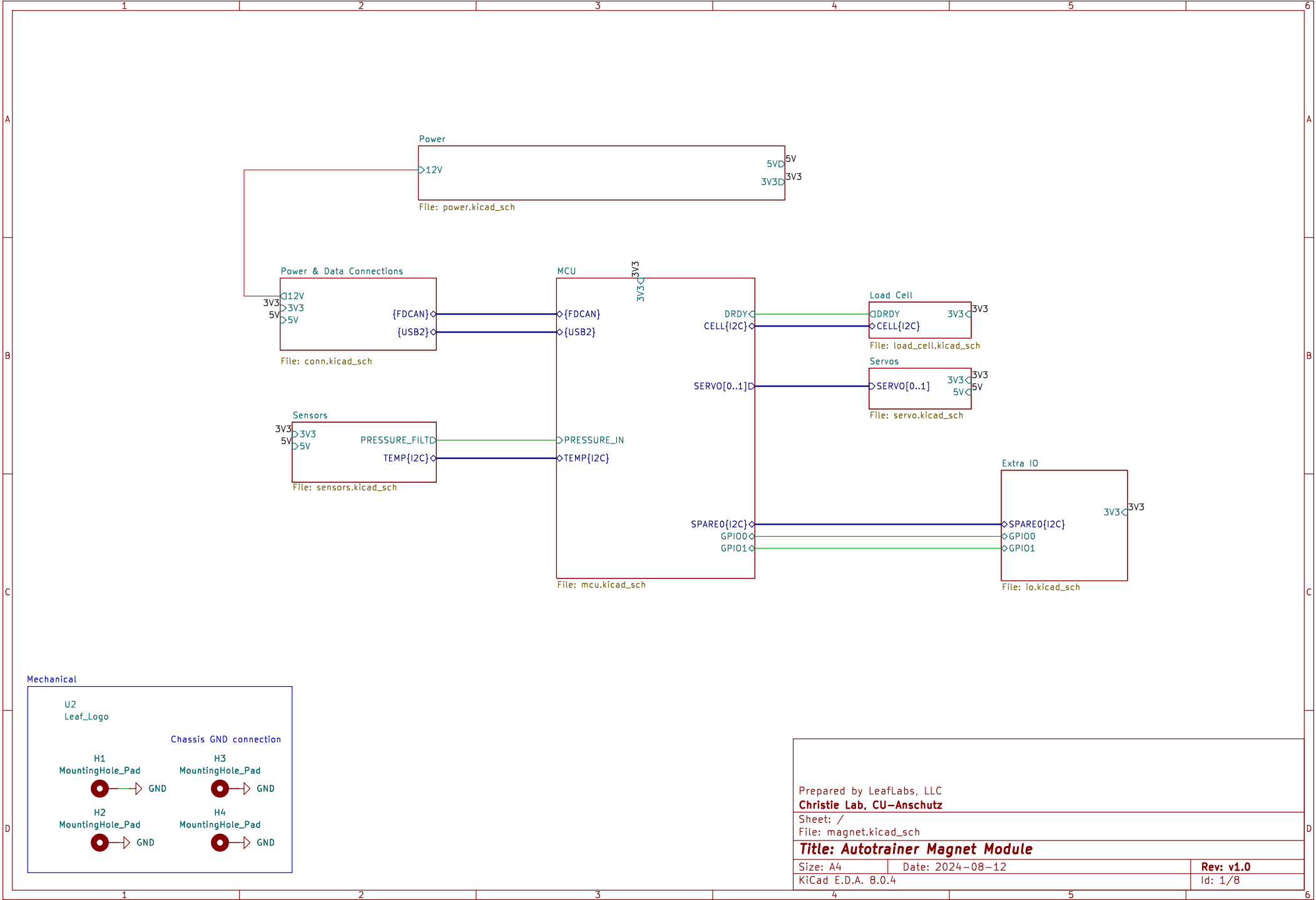
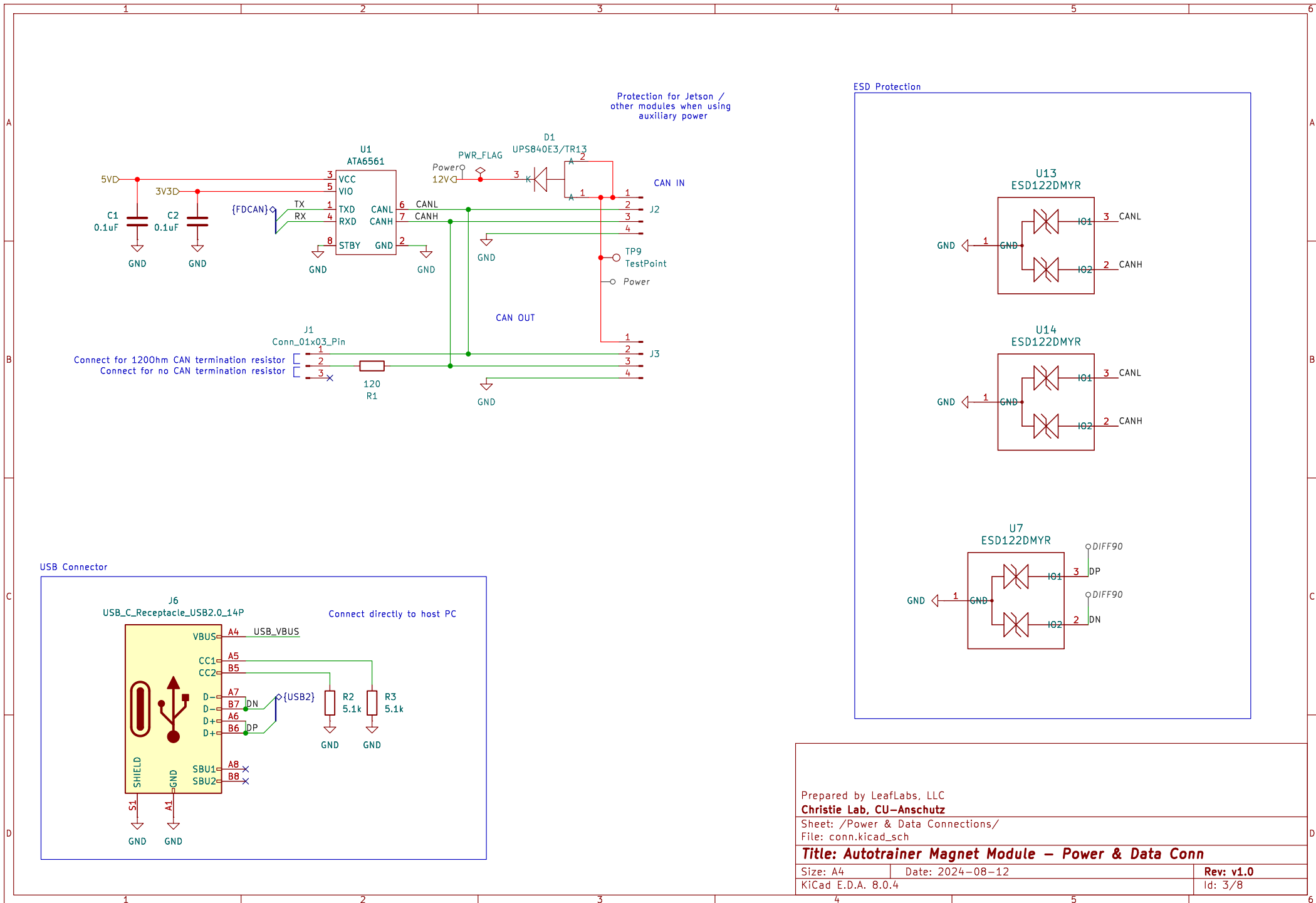
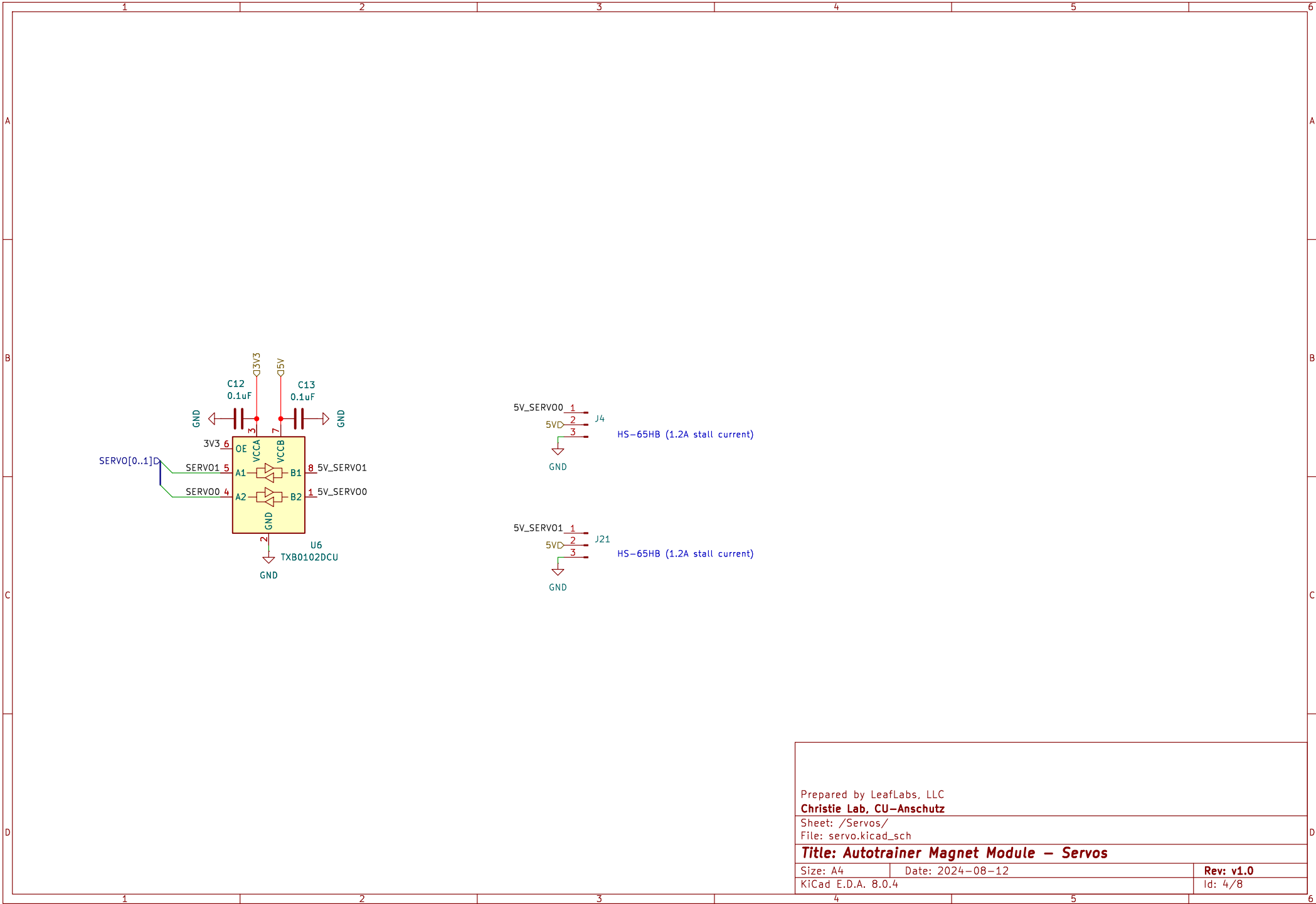








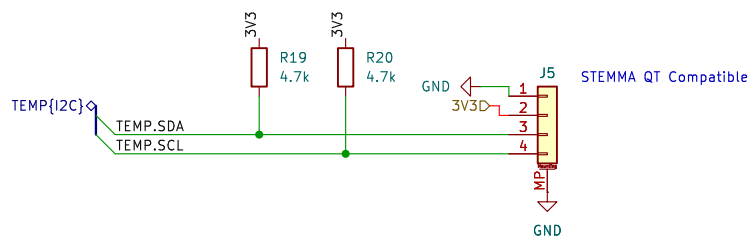
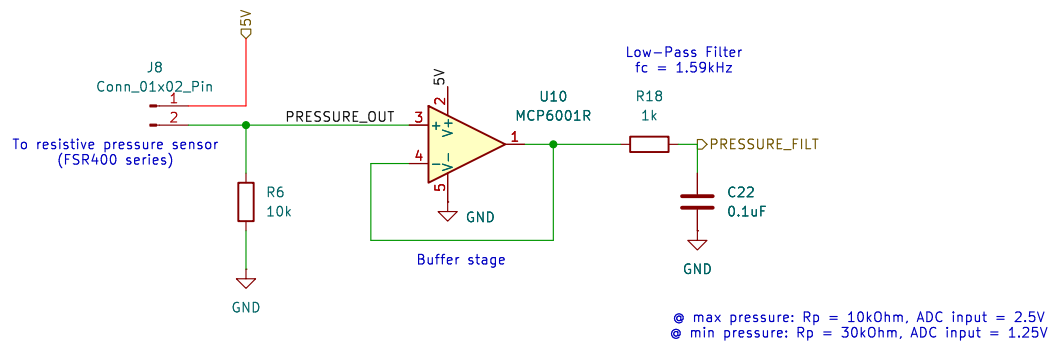
Prepared by LeafLabs, LLC
Christie Lab, CU-Anschutz

Sheet: /Servos/
File: servo.kicad_sch

Title: Autotrainer Magnet Module – Servos

Size: A4	Date: 2024-08-12	Rev: v1.0
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Sheet: /Sensors/
File: sensors.kicad_sch

Title: Autotrainer Magnet Module – Sensors

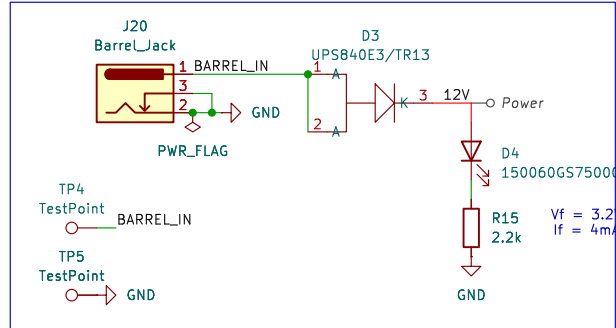
Size: A4	Date: 2024-08-12
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Size: A4	
KiCad E.D.A. 8.0.4	

Rev: v1.0

Id: 7/8

Aux Power In



12V to 5V converter
Max expected current consumption: 2.7 A

Inductor selection math

$$I_{pp} = V_{out}/V_{in(max)} * (V_{in(max)} - V_{out})/(L_o * f_{sw})$$

$$I_{pp} = 5/12.1 * (12.1 - 5)/(3.3\mu H * 550 \text{ kHz})$$

$$I_{pp} = 1.6 \text{ A}$$

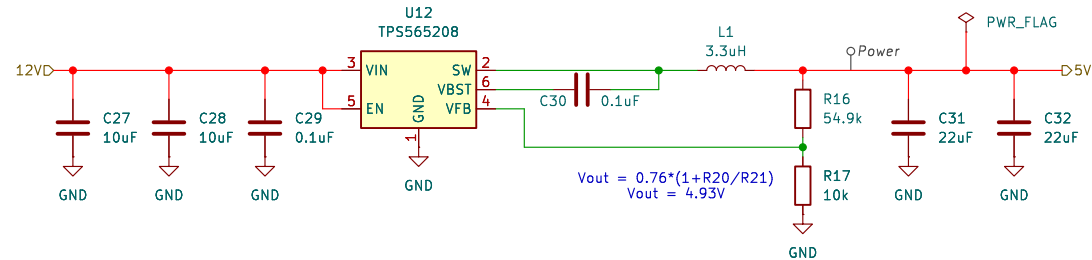
$$I_{peak} = I_o + I_{pp}/2 = 2.7 + 0.8 = 3.5 \text{ A}$$

$$I_o(rms) = \sqrt{I_o^2 + 1/12 I_{pp}^2} = 2.74 \text{ A}$$

Capacitor current

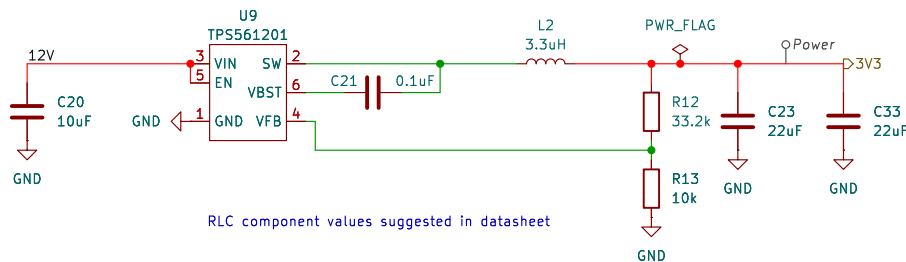
$$I_{co(rms)} = (V_{out} * (V_{in} - V_{out})) / (\sqrt{12} * V_{in} * L * f_{sw})$$

$$I_{co(rms)} = 1.11 \text{ A}$$



12V to 3V3 Converter

Expected current consumption: 126mA
Supports max 500mA

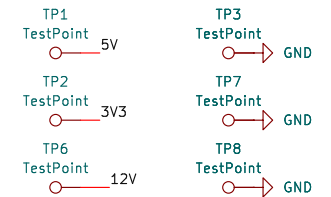


RLC component values suggested in datasheet

$$I_{pp} = 3.3/12.1 * (12.1 - 3.3)/(3.3\mu H * 580\text{kHz}) = 1.25 \text{ A}$$

$$I_{peak} = 0.5 + 1.25/2 = 1.125 \text{ A}$$

$$I_{lo(rms)} = \sqrt{0.5^2 + 1/12 I_{pp}^2} = 0.62 \text{ A}$$



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Sheet: /Power/
File: power.kicad_sch

Title: Autotrainer Magnet Module – Power

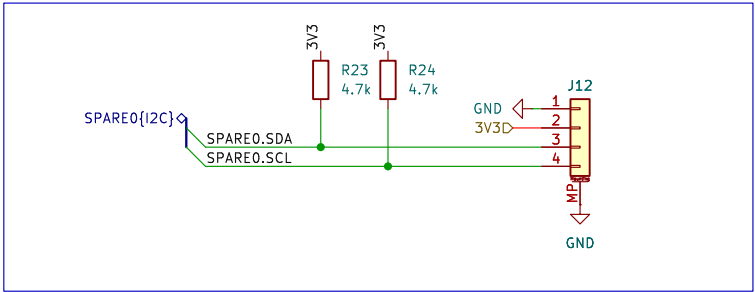
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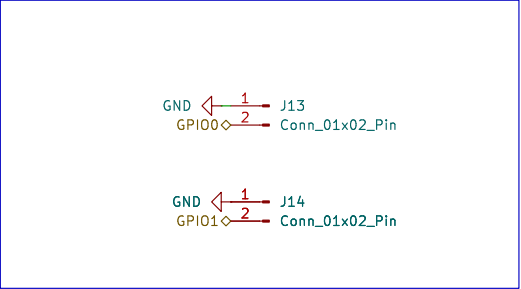
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STEMMA QT Connector



Extra GPIO



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Sheet: /Extra IO/
File: io.kicad_sch

Title: Autotrainer Magnet Module – External IO

Size: A4	Date: 2024-08-12	Rev: v1.0
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